

Predicates and Quantifiers

Gaps in Propositional Logic

Suppose we know that

"Every computer on the university network is functioning properly"

No rules of propositional logic to conclude

"MATH3 is functioning properly"



Suppose we know that

"CS2 does not functioning properly"

No rules of propositional logic to conclude

"There is a computer on the university network that does not work properly"



Propositional Function / Predicate

A **Predicate** is a declarative sentence whose true/false value depends on one or more variables.

Denote the statement “ x is greater than 3” by $P(x)$.

- Note that it is not a proposition.
- $P(4)$ = “4 is greater than 3” is a proposition, its value is *true*
- $P(2)$ = “2 is greater than 3” is a proposition, its value is *false*
- so P takes a number and sends it to a proposition (either true or false).
- P is a “function” from numbers to propositions.
- P is also called a **propositional function**
- The part “...greater than 3” of $P(x)$ can be viewed as predicate, x is the variable.

$$P: (\text{num}) \rightarrow \{0, 1\}$$

$= Q(x, y)$ true false

Statement " $x = y + 3$ " involves now two variables. Denote this state by $Q(x, y)$. What are the values of $Q(4, 1)$, $Q(2, 1)$?

Here propositional function Q (predicate) is of two variables and formally is

$\swarrow \searrow$ numbers
" = + 3."

A statement of the form $P(x_1, x_2, \dots, x_n)$ is the value of the propositional function P at the n -tuple $(x_1, x_2, \dots, x_n)$, and P is also called an n -place predicate or a n -ary predicate.

$$|x| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} = |x| > 0$$

Example

Given a vector $\mathbf{x} = (x_1, x_2, \dots, x_n)$ its magnitude is positive $\|\mathbf{x}\| > 0$.

Denote that statement by $P(x_1, x_2, \dots, x_n)$. This is not a proposition (since $P(\mathbf{0})$ is false and $P(\mathbf{x})$ is true for any non-zero vector $\mathbf{x}$. It is an n -ary predicate.

Examples

Given each propositional function determine its true/false value when variables are set as below.

1) $Prime(x) = \text{"x is a prime number."}$

$Prime(2)$ is *true*

$Prime(9)$ is *false*

2) $C(x, y) = \text{"x is the capital of y"}$.

$C(\text{Ottawa}, \text{Canada})$ is *true*

$C(\text{Buenos Aires}, \text{Brazil})$ is *false*

3) $E(x, y, z) = \text{"}x^2 + y^3 = z^5\text{"}$.

$E(2, 3, 5)$ is *true*

$E(4, 4, 17)$ is *false*

Answer Q1

Predicates in computer programs

condit. oper.

Consider the statement

If $x > 0$ **then** $x := x + 1$

The predicate here is

$P(x) = "x > 0"$

What is the output if
the input is $x = -1$?

$x = -1$

What is the output if
the input is $x = 3$?

$x = 4$

Answer Q2

Let $x := 3$ and $y := 5$

Let $P(a, b) \equiv "x = a \text{ and } y = b"$

$P(3, 5)$ is $"x=3 \text{ and } y=5" = \text{true}$

$P(5, 3)$ is $"x=5 \text{ and } y=3" = \text{false}$

$temp := x$

$x := y$

$y := temp$

Now $P(5, 3)$ is true

We can check if the blue part of
a computer program is correct

If $P(5, 3)$ **then** "correct"

Quantifiers

Let $P(x) = "x \text{ is odd}"$

Construction of a proposition from a predicate:

① Evaluate the predicate on some value; $P(7)$ is true, $P(2)$ is false.

② **Quantification** is another way to create proposition from a predicate.

"There is x such that x is odd" - this is now a proposition, since it has truth value TRUE. *"TRUE so"*

- In English, the words **all**, **some**, **many**, **none**, and **few** are used.
- **Universal quantification**, $\forall x P(x)$ says
"predicate P is true for every element under consideration."
Under the domain of natural numbers, $\forall x P(x)$ is *false*
- **Existential quantification** $\exists x P(x)$ says
"There is at least one element under consideration for which the predicate P is true."
Under the domain of natural numbers, $\exists x P(x)$ is *true*
- The area of logic that deals with predicates and quantifiers is called the **predicate calculus**.

Domain, domain of discourse, universe of discourse

$\forall xP(x)$ = "For all x we have that x is odd" TRUE or FALSE?

We cannot decide without domain !!!

$\forall xP(x)$ for the domain of **integer numbers** is *false*
but $\forall xP(x)$ for the domain of **prime numbers greater than 2** is *true*

$\exists xP(x)$ = "There is x such that x is odd" TRUE or FALSE?

$\exists xP(x)$ for the domain of **all prime numbers** is *true*
but $\exists xP(x)$ for the domain of **all even numbers** is *false*

Answer Q3

Universal quantifier

The universal quantification of $P(x)$ is the statement

“ $P(x)$ for all values of x in the domain.”

The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$.

Here $\forall$ is called the **universal quantifier**.

$\forall x P(x)$ is read as “for all x , $P(x)$ ” or “for every x $P(x)$.”

$\forall x P(x)$ is true when $P(x)$ is true for every x in the domain (proof).

$\forall x P(x)$ is false when there is an x in the domain for which $P(x)$ is false.

An element for which $P(x)$ is false is called a **counterexample** of $\forall x P(x)$.

Let $P(x)$ be the statement “ $x^2 + x + 1 > 0$ ” and the domain all real numbers.

$\forall x P(x)$ is **true**

proof: Solve $x^2 + x + 1 = 0$, $D = 1 - 4 < 0 \Rightarrow$ no sol.
 $\Rightarrow y = x^2 + x + 1$ never meets the x -axis, so $y > 0$

Let $P(x)$ be the statement “ $x < 1$ ” and the domain all real numbers.

$\forall x P(x)$ is FALSE.

A counterexample is **$x = 2$**

Answer Q4

Universal quantifier with final domain

When all the elements in the domain can be listed as $x_1, x_2, \dots, x_n$ then $\forall x P(x)$ is the same as $P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$.

Example: What is the truth value of $\forall x P(x)$, where $P(x)$ is " $x^2 < 10$ " and the domain consists of the positive integer not exceeding 4. That is domain is $\{1, 2, 3, 4\}$.

$\forall x P(x)$ is true in the only case when $P(1), P(2), P(3)$ and $P(4)$ are true at the same time.

Hence

$$\forall x P(x) \equiv P(1) \wedge P(2) \wedge P(3) \wedge P(4) = 0$$

$x=4$ is count.ex.

Existential Quantifier

The existential quantification of $P(x)$ is the proposition

"There exists an element x in the domain such that $P(x)$."

Denoted as $\exists x P(x)$

Here $\exists$ is called the *existential quantifier*.

Reads as "There is x , $P(x)$ " or "There is at least one x , $P(x)$."

$\exists x P(x)$ is true when $P(x)$ is true for one or more x in the domain.

An element for which $P(x)$ is true is called a **witness** of $\exists x P(x)$.

$\exists x P(x)$ is false when $P(x)$ is false for every x in the domain (proof).

In the domain of integers $\exists x(x > 3)$ is *true* and the witness can be $x=4$

In the same domain $\exists x(x = x + 1)$ is *false*

proof: $x = x + 1$ is $0 = 1$ which is false for any x .

Uniqueness quantifier: "There exists a **unique** x such that $P(x)$ " = $\exists! x P(x)$.

$\exists! x(x + 1 = 2)$ is *true* ($x=1$)

Answer Q5

Existential quantifier with final domain

When all the elements in the domain can be listed as $x_1, x_2, \dots, x_n$ then $\exists x P(x)$ is the same as $P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$.

Example: What is the truth value of $\exists x P(x)$, where $P(x)$ is " $x^2 < 10$ " and the domain consists of the positive integer not exceeding 4.

That is domain is $\{1, 2, 3, 4\}$.

$\exists x P(x)$ is true when at least one of $P(1), P(2), P(3)$ and $P(4)$ is true.

Hence

$$\exists x P(x) \equiv \overset{=1}{P(1)} \vee \overset{=1}{P(2)} \vee \overset{=1}{P(3)} \vee \overset{=0}{P(4)} = 1$$

witness is $x = 1$ (or $x = 2$ or $x = 3$)

Answer Q6

Restricting the domain of a quantifier

$\forall x > 0 (x^2 > 0)$ is the same as $\forall x (x > 0 \Rightarrow x^2 > 0)$.
 $\exists y > 0 (y^2 = 2)$ is the same as $\exists y (y > 0 \wedge y^2 = 2)$.

remember!!!

Precedence of quantifier

$\forall$ and $\exists$ have higher precedence than logical operators.

$\forall x P(x) \vee Q(x)$ means $(\forall x P(x)) \vee Q(x)$, it does NOT mean $\forall x (P(x) \vee Q(x))$.

Binding variables

When a quantifier is used on the variable x we say that x is **bound**.

When a variable is not bound by a quantifier or set to a particular value, the variable is said to be **free**.

$\exists x (x + y = 1)$ x is bound by $\exists$, but y is free

Scope

The part of a logical expression to which a quantifier is applied is called the **scope** of this quantifier.

In $\exists x (P(x) \wedge Q(x)) \vee \forall y R(y)$, the scope of $\exists x$ is $P(x) \wedge Q(x)$ and the scope of $\forall y$ is $R(y)$.

Definition

Two statements S and T involving predicates and quantifiers are **logically equivalent** if

- they have the same truth value!
- no matter which predicates are substituted into these statements and
- no matter which domain of discourse is used for the variables in these propositional functions.
- We use the notation $S \equiv T$

$$\forall x(P(x) \wedge Q(x)) \equiv \forall xP(x) \wedge \forall xQ(x).$$

Sketch of Proof:

→ Let $\forall x(P(x) \wedge Q(x))$ is true then for any a from domain we have that $P(a) \wedge Q(a) \equiv 1 \Rightarrow P(a) \equiv 1$ and $Q(a) \equiv 1$, hence, $\forall xP(x) \equiv 1$ and $\forall xQ(x) \equiv 1 \Rightarrow \forall xP(x) \wedge \forall xQ(x) \equiv 1$
← EX.

$$\forall x(P(x) \vee Q(x)) \not\equiv \forall xP(x) \vee \forall xQ(x)$$

find one example of P, Q , domain, where we have $\neq$
 $P(x) = "x \text{ is male}"$ } for any person, he/she is male or female $\equiv 1$
 $Q(x) = "x \text{ is female}"$ } any person is male or any person is female $\equiv 0$
domain - all people

Negating Quantifiers Expressions

Let $\forall xP(x)$ = "Every student in your class has taken Calculus"

Negation of it

$\neg\forall xP(x)$ = "It is not the case that every student in your class has taken Calculus"

or

$\neg\forall xP(x)$ = "THERE IS a student in your class who has NOT taken Calculus"

↓

$$\neg\forall xP(x) \equiv \exists x\neg P(x) \leftarrow \text{learn!}$$

Let $\exists xP(x)$ = "There is a student in your class who has taken Calculus"

Negation of it

$\neg\exists xP(x)$ = "It is not the case that there is a student in your class who has taken Calculus" or

$\neg\exists xP(x)$ = "Every student in your class has not taken Calculus"

$$\neg\exists xP(x) \equiv \forall x\neg P(x) \leftarrow \text{learn!}$$

These two Laws are called **De Morgan's Laws**. WHY?

Why it is called De Morgan's Law?

Assume domain is finite, that is $D = \{x_1, x_2, \dots, x_n\}$. Then

$$\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n).$$

By De Morgan's Law

$$\neg (P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)) = \neg P(x_1) \vee \neg P(x_2) \vee \dots \vee \neg P(x_n).$$

However,

$$\exists x Q(x) \equiv Q(x_1) \vee Q(x_2) \vee \dots \vee Q(x_n). \quad \text{Handwritten: } Q = \neg P$$

Therefore, $\neg (\forall x P(x)) \equiv \exists x \neg P(x)$.

For the second law, substitute $Q = \neg P$ in the first law to get

$\neg (\forall x \neg Q(x)) \equiv \exists x Q(x)$ and apply negation. One gets $\forall x \neg Q(x) \equiv \neg (\exists x Q(x))$.

$$\text{Handwritten: } \neg (\forall x \neg Q) \equiv \neg (\exists x Q(x))$$

Use negation step-by-step

$$\neg(\forall x(P(x) \rightarrow Q(x))) \equiv \exists x \neg(P \rightarrow Q) \equiv \exists x \neg(\neg P \vee Q) \equiv \exists x (\neg\neg P \wedge \neg Q) \equiv \exists x (P \wedge \neg Q)$$

Negate the following:

- 1 There is an honest politician.
- 2 All Americans eat cheeseburgers.
- 3 $\exists x(x^2 = 2)$ (what is the truth value?)

Answer:

- 1 $\neg(\exists x H(x)) \equiv \forall x \neg H(x)$, that is “All politicians are dishonest”.
- 2 $\neg(\forall x C(x)) \equiv \exists x \neg C(x)$, “there is an American who doesn’t eat cheeseburgers”.
- 3 $\neg(\exists x(x^2 = 2)) \equiv \forall x(x^2 \neq 2)$ which is false if domain is real numbers, and true if the domain is integers.

Exercises:

$$\begin{aligned}\exists x(P(x) \vee Q(x)) &\equiv \exists x P(x) \vee \exists x Q(x) \\ \exists x(P(x) \wedge Q(x)) &\not\equiv \exists x P(x) \wedge \exists x Q(x)\end{aligned}$$

Translate into logic

Assume $P(x) = "x \text{ is perfect}"$ and $F(x) = "x \text{ is your friend}"$.

a) No one is perfect

$$\forall x \neg P(x) \equiv \neg(\exists x P(x))$$

b) Not everyone is perfect

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x)$$

c) All your friends are perfect

↙ domain

$$\forall x (F(x) \rightarrow P(x))$$

d) At least one of your friends is perfect

$$\exists x (F(x) \wedge P(x))$$

e) Everyone is your friend and ^{everyone} is perfect

$$\forall x (F(x) \wedge P(x)) \equiv \forall x F(x) \wedge \forall x P(x)$$

f) Not everybody is your friend or someone is not perfect

$$\begin{aligned} & \neg \forall x F(x) \vee \exists y \neg P(x) \equiv \\ & \equiv \exists x \neg F(x) \vee \exists y \neg P(x) \end{aligned}$$

Answer Q7

Quantifiers in System Specifications

“Every mail message larger than 1 Mb will be compressed.”

$S(m, y)$ = “Mail message m is larger than y Mb”

$C(m)$ = “Mail message m will be compressed”

Domains for m is all mail messages y is a positive real number.

$\forall m(S(m, 1) \rightarrow C(m))$.

“If a user is active, at least one network link will be available”

$A(u)$ = “User u is active”

Domains for u is all users.

$S(n, x)$ = “Network link n is in state x ”

Domains for n is all network links and x can be available or not.

$\exists u A(u) \rightarrow \exists n S(n, available)$.

Homework:

- p. 51 of the book read the paragraph “Logic Programming”
- Solve all *odd* exercises starting at p. 53.

Nested Quantifiers

Understanding statements

Two quantifiers are **nested** if one is in the scope of the other.

Everything within the scope of a quantifier can be thought of as a propositional function.

For example,

$\forall x \exists y (x + y = 0)$ is the same as $\forall x Q(x)$,
where $Q(x)$ is $\exists y (x + y = 0)$.

$\forall x \exists y (x + y = 0) \quad (y = -x) \equiv 1$	for every real number x there is a real number y such that $x + y = 0$
$\forall x \forall y (x + y = y + x) \equiv 1$	$x + y = y + x$ for all reals x and y
$\forall x \forall y \forall z (x + (y + z) = (x + y) + z)$	the associativity law for real numbers
$\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0)) \equiv 1$	The product of a positive real number and a negative real number is always a negative real number.

Answer Q8

The order of quantifiers

$$\forall x \forall y Q(x, y) \equiv \forall y \forall x Q(x, y)$$

$$\forall x \forall y (x + y = y + x) = \forall y \forall x (x + y = y + x)$$

$$\forall x \exists y P(x, y) \not\equiv \exists y \forall x P(x, y)$$

$\forall x \exists y (x + y = 0)$ is *true* ($y = -x$)
 $\exists y \forall x (x + y = 0)$ is *false*

Let $Q(x, y, z)$ be the statement " $x + y = z$ " (a ternary predicate).

What are the truth values of $\forall x \forall y \overset{=x+y}{\exists z} Q(x, y, z)$ and $\exists z \forall x \forall y Q(x, y, z)$?

Answers:

true

false

Answer Q9

Mathematical statements into Logic

Translate the statement "The sum of two positive integers is always positive" into a logical expression.

$$\forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow x + y > 0)$$

Translate the statement

"Every real number except zero has a multiplicative inverse." (A multiplicative inverse of a real number x is a real number y such that $xy = 1$.)

$$2 \cdot \frac{1}{2} = 1$$

$$\forall x ((x \neq 0) \rightarrow \exists y (xy = 1))$$

$$\lim_{x \rightarrow a} f(x) = L$$

For every real number $\forall \varepsilon > 0$ there exists a real number $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta$.

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x ((0 < |x - a| < \delta) \rightarrow |f(x) - L| < \varepsilon)$$

Answer Q10

Translate from logic into English

Domain is the students of IUT, $C(x) :=$ student x has a computer,
 $F(x, y) =$ students x and y are friends,

$$\forall x(C(x) \vee \exists y(C(y) \wedge F(x, y)))$$

Answer: Any student of IUT has a computer or he has a friend who has a computer.

Domain is the students of IUT, $F(x, y) =$ students x and y are friends,

$$\exists x \forall y \forall z ((F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z))$$

Answer: There is a student in IUT whose any two ^{different} friends are not friends between each other.

Answer Q11

Translate from English into logic:

$F(x)$

$P(x)$

$M(x,y) = x$ is mother of y

If a person is female and is a parent, then this person is someone's mother.

$$\forall x ((F(x) \wedge P(x)) \rightarrow \exists y M(x,y))$$

Everyone has exactly one best friend. $F(x,y) = x$ is best friend of y

$$\forall x \exists! y F(y,x) \equiv \forall x (\exists y F(y,x) \wedge \forall z \neq y \neg F(z,x))$$

There is a woman who has taken a flight on every airline in the world.

$Q(w, f, a) = "w \text{ has taken } f \text{ on } a"$

$$\exists w \exists f \forall a Q(w, f, a) \leftarrow \text{wrong}$$

$$\exists w \forall a \exists f Q(w, f, a) \leftarrow \text{correct}$$

Answer Q12

Negating nested quantifiers

- $\forall x \exists y (xy = 1)$

Answer: $\exists x \forall y (xy \neq 1)$

$$\neg(\forall x \exists y (xy = 1)) = \exists x \neg(\exists y (xy = 1)) = \exists x \forall y \neg(xy = 1) \\ \rightarrow = \exists x \forall y (xy \neq 1)$$

- The limit of a function f at the point $x = a$ exists.

Solution: The limit does not exist if for any real number L one has

$\lim_{x \rightarrow a} f(x) \neq L$, which is $\neg(\lim_{x \rightarrow a} f(x) = L)$.

$$\neg(\forall \varepsilon > 0 \exists \delta > 0 \forall x ((0 < |x-a| < \delta) \rightarrow |f(x) - L| < \varepsilon)) \equiv \\ \equiv \exists \varepsilon > 0 \forall \delta > 0 \exists x ((0 < |x-a| < \delta) \wedge |f(x) - L| \geq \varepsilon) \\ \equiv \exists \varepsilon > 0 \forall \delta > 0 \exists x ((0 < |x-a| < \delta) \wedge |f(x) - L| \geq \varepsilon)$$

Answer Q13

Homework:

- Solve *odd* exercises starting at p. 64, #1-#47.